

NER-SMARTLOG | Hackathon Executive Summary & Judges Q&A Guide

Smart India Hackathon 2026 - Problem Statement 26002 | Ministry of Development of NER [MDoNER]

1. EXECUTIVE PROJECT SUMMARY

NER-SMARTLOG is an AI-powered predictive logistics and road hazard telemetry portal engineered for the 8 North Eastern states (Assam, Arunachal Pradesh, Meghalaya, Manipur, Mizoram, Nagaland, Tripura, and Sikkim). It resolves critical supply chain bottlenecks (vaccines, oxygen, food relief) during severe monsoon landslides and floods.

KEY SYSTEM INNOVATIONS:

- * Predictive Multi-Corridor AI Router: Computes and ranks alternative mountain detours against a dynamic Risk Index.
- * Real-Time 8-State Weather Telemetry: Live Open-Meteo NWP sensor integration for all NER states and Himalayan passes.
- * Multi-Role Command Matrix: Specialized interfaces for Ground Patrol (BRO), Logistics Commanders, & SDMA Disaster Cells.
- * NER-SMARTBOT AI Assistant: Instant natural language copilot for road passability and convoy status.

MATHEMATICAL RISK SCORING FORMULA:

Risk Score = (0.30 x Rain) + (0.25 x Landslide) + (0.20 x Road_Status) + (0.10 x Traffic) + (0.10 x History) + (0.05 x Elevation)

- 0 to 30 : LOW RISK (Highway Open & Clear)
 - 31 to 60 : MEDIUM RISK (Single-lane Caution / Minor Mudflow)
 - 61 to 80 : HIGH RISK (Active Hazard / Rockfall Warning)
 - 81 to 100 : CRITICAL RISK (Road Blocked - Automated AI Detour Reroute Triggered)
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2. TOP HACKATHON JUDGING QUESTIONS & WINNING ANSWERS

Q1. What is the core innovation of NER-SMARTLOG over Google Maps?

Ans: Commercial navigation apps optimize strictly for shortest driving time based on current traffic. In the Himalayas, roads look clear until a sudden landslide strikes. NER-SMARTLOG evaluates predictive geological & meteorological risk (rainfall rate, slope gradient, verified BRO alerts) to proactively reroute relief convoys before they enter hazard zones.

Q2. How does the system operate in remote mountain areas with zero internet connectivity?

Ans: Convoys use an Offline-First architecture with pre-cached vector maps. In cellular black zones, the telemetry unit falls back to SMS/USSD compressed encrypted status packets to maintain grid connectivity.

Q3. Where is the live weather telemetry sourced from?

Ans: Directly from Open-Meteo High-Resolution Numerical Weather Prediction (NWP) API across all 8 state nodes.

Q4. Explain the routing algorithm. How do you generate alternative corridors?

Risk Score formula is applied. If the primary highway exceeds Risk > 60, the AI automatically selects the safest detour.

Q5. How does the system ensure authentic disaster reports?

Ans: We enforce Role-Based Access Control (RBAC). Only verified personnel with Department Badge IDs (BRO, NDRF, SDMA) have broadcast privileges, preventing false civilian alarms.

3. 60-SECOND ELEVATOR PITCH FOR JUDGES

"Respected Judges, in North East India, a landslide doesn't just block a road - it cuts off life-saving oxygen, vaccines, and emergency food supplies to remote border communities. Commercial navigation only reacts after a road is blocked. NER-SMARTLOG is the first predictive street logistics and hazard telemetry portal built specifically for MDoNER. By fusing live weather sensor feeds, geological risk scoring, and verified ground BRO telemetry, our AI engine detects impending corridor failures and re-routes relief convoys before they get stranded."